

FORM 2 THE PATENTS ACT 1970 39 OF 1970 & THE PATENT RULES 2003 COMPLETE SPECIFICATION (SEE SECTIONS 10 & RULE 13)		
1. TITLE OF THE INVENTION “ SYSTEM FOR SOLAR CELL ELECTRICAL CHARACTERISTICS MEASUREMENT”		
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3. PREAMBLE TO THE DESCRIPTION		
COMPLETE SPECIFICATION The following specification particularly describes the invention and the manner in which it is to be performed.		

TECHNICAL FIELD

[001] The present invention relates to a system for solar cell electrical characteristics measurement. Particularly, the present invention relates to a system
5 which includes a light mixer, a solar cell, and a probing unit for determining the electrical characteristics of solar cells under simulated sunlight.

BACKGROUND OF THE INVENTION

[002] The background description includes information that may be useful in
10 understanding the present invention.

[003] Accurate evaluation of solar cell performance necessitates both the simulation of sunlight and the precise measurement of electrical characteristics. This evaluation process generally involves two primary components: a solar simulator to generate controlled illumination and a probing mechanism to establish electrical
15 contact for measurement purposes. Initial consideration will be given to solar simulators employed within this domain.

[004] Xenon Solar Simulators are extensively utilized in the field of solar testing and simulation due to their capability to generate a broad spectrum of light that closely approximates natural sunlight. These simulators function by employing a
20 Xenon arc lamp that emits a continuous spectrum encompassing UV, visible, and near-IR regions. To modulate the spectrum and attain a desired output, optical filters are integrated. Band-pass filters are implemented to selectively permit specific wavelength ranges to pass while attenuating others, and neutral density filters serve to adjust the intensity of the light. However, despite their demonstrated
25 effectiveness, Xenon simulators are susceptible to limitations imposed by inherent light absorption and scattering within the optical filters. This inherent characteristic can lead to a reduction in the overall intensity and efficiency of the simulated sunlight.

[005] LED-Based Solar Simulators have risen to prominence as a more adaptable and cost-effective alternative when compared to Xenon-based systems. These
30 simulators operate through the utilization of LED arrays, with each LED emitting light at discrete wavelengths to collectively address different segments of the solar spectrum. Spectral output is meticulously adjusted through the selection of LEDs exhibiting appropriate emission spectra to closely replicate the desired sunlight

simulation. Light mixing within LED-based simulators is frequently achieved via optical diffusion techniques, such as diffusers or mixing chambers, designed to homogenize the light output from the array of LEDs to produce a more uniform illumination. It should be noted, however, that these diffusion methodologies can
5 also induce light absorption or scattering, potentially influencing both the intensity and uniformity of the simulated spectrum.

[006] Existing systems have employed methodologies to establish electrical contact with the solar cell and quantify its performance. Traditional probing mechanisms are of critical importance in establishing this electrical connection. A prevalent and
10 widely adopted approach for probing mechanisms in solar cell testing involves the utilization of gold-plated busbars in conjunction with pogo pins to establish electrical contact with the solar cell. This approach offers both flexibility and reliable connectivity, as pogo pins, engineered with spring-loaded contacts, ensure consistent pressure distribution across the cell's surface. Notwithstanding its widespread
15 adoption, this methodology is subject to several inherent limitations that can compromise the accuracy of current-voltage (IV) curve measurements.

[007] A primary concern associated with pogo pin systems is the potential for creation of a potential gradient across the pogo pins during testing procedures. As the pins establish electrical contact at disparate locations on the cell's busbar,
20 variations in electrical resistance can precipitate uneven current distribution. This non-uniformity in current flow can result in inaccuracies during the measurement of the true IV characteristics of the solar cell, consequently leading to potential distortions of key performance indicators, such as open-circuit voltage (Voc), short-circuit current (Isc), and the fill factor (FF).

[008] Moreover, the physical configuration of the probing mechanism, including the vertical dimension and spatial arrangement of busbars and pogo pins, can induce shadowing effects on the solar cell during measurement acquisition. This shading phenomenon diminishes the quantity of light incident upon the active area of the cell, which can directly and adversely affect the fill factor and the overall efficiency
25 values derived from the IV curve. Since shading fundamentally alters the uniformity of light exposure, it inherently compromises the reliability of performance assessments, particularly when evaluating parameters such as maximum power point (MPP).
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[009] To reiterate salient points, pogo pin systems, while offering adaptability, inherently present challenges owing to their physical structure. As an example, stemming from the physical height characteristic of pogo pin structures, shading inevitably occurs when light emanating from the solar simulator is partially obstructed by components of the probing system. This obstruction reduces the effective irradiance incident upon the solar cell. This shading effect consequently exerts a negative impact on the fill factor and has the potential to introduce inaccuracies into the measurement of the solar cell's overall efficiency.

[010] Beyond considerations of shading, pogo pin systems also have the potential to introduce potential gradients across the busbars during IV measurements. When multiple pins establish contact with the solar cell at various locations, minor fluctuations in electrical resistance between contact points can instigate a non-uniform distribution of current. This, in turn, leads to distorted voltage readings, resulting in errors within the derived IV curve and potential underestimation or overestimation of the cell's performance parameters. The inherent inconsistency in contact resistance becomes particularly problematic in high-precision testing environments, where even minor inaccuracies can significantly impact the interpretation of the solar cell's intrinsic quality and performance characteristics under varying operational conditions.

[011] In response to certain limitations associated with pogo pin methodologies, alternative approaches have been investigated. One such alternative method involves the utilization of fixed probes or mechanical arms equipped with conductive pads or gold-plated contacts, engineered to establish either a single point or line contact with the cell. While this methodology can mitigate the risk of uneven current distribution in comparison to pogo pin systems, it nonetheless remains susceptible to challenges related to shading and mechanical wear. Fixed probes frequently necessitate precise alignment to preclude the application of excessive force on the cell, which could potentially induce damage or result in diminished contact reliability over extended periods. These systems, similar to pogo pin configurations, also require appreciable vertical clearance, which, again, contributes to the potential for shadowing on the cell's surface.

[012] In a further effort to enhance contact uniformity, Vacuum-based contact systems are implemented to draw the solar cell into a contact fixture, thereby improving the consistency of contact between the solar cell and the busbars. While

this approach can effectively address certain contact resistance concerns, it introduces added complexity into the overall system architecture and may prove impractical for larger-scale testing scenarios or for the manipulation of delicate solar cells, such as those employing thin-film technologies.

5 **[013]** In summary, existing systems present critical gaps in accurately simulating sunlight both with high intensity and spectral uniformity, leading to potential inaccuracies in performance assessments. Furthermore, traditional probing mechanisms, such as pogo pin systems, introduce significant challenges due to shadowing and the creation of potential gradients across the solar cell, directly
10 compromising the reliability of electrical characteristic measurements, particularly for crucial parameters like fill factor and IV curves.

[014] Therefore, there is a pressing need to address these gaps, a system that would necessitate a light mixer capable of generating highly uniform and intense simulated sunlight, and a probing mechanism designed to minimize shadowing and ensure
15 consistent, accurate electrical contact along the solar cell busbars.

OBJECTS OF THE INVENTION

[015] Some of the objects of the present disclosure, which at least one embodiment herein satisfy, are listed herein below.

20 **[016]** The object of the present system is for measuring the electrical characteristics of solar cells using simulated sunlight.

[017] Another object of the present system is to provide a light mixer that efficiently simulates sunlight with high intensity and uniformity, contributing to the accuracy and reliability of solar cell electrical characteristic measurements.

25 **[018]** Yet another object of the present system is to ensure gentle and uniform contact pressure between the probing mechanism and the solar cell busbars, preventing damage to the cell while ensuring consistent electrical connection.

[019] Yet another object of the present system is to provide a probing mechanism that facilitates more accurate and reliable current-voltage (IV) curve measurements,
30 leading to a more precise characterization of solar cell performance.

[020] Yet another object of the present system is for measuring the electrical characteristics of both monofacial and bifacial solar cells, optimizing testing configurations for each cell type.

[021] Yet another object of the present system is to minimize shadowing by the probing mechanism on the solar cell during testing, leading to more uniform light exposure and consequently, more accurate measurement of solar cell electrical characteristics.

- 5 [022] These and other objects and advantages will become more apparent when reference is made to the following description and accompanying drawings.

SUMMARY OF THE INVENTION

[023] This summary is provided to introduce the concepts related to a system
10 for solar cell electrical characteristics measurement.

[024] In an aspect of the present disclosure, there is disclosed a system for solar cell electrical characteristics measurement. The system includes a light mixer, a solar cell, and a probing unit. The light mixer is configured to simulate sunlight illumination. Further, the solar cell is configured to receive the simulated
15 sunlight. Furthermore, the probing unit is configured for probing the solar cell for electrical characteristics measurement.

[025] In another aspect of the present disclosure, said solar cell may be one of monofacial cell and bifacial cell.

[026] In another aspect of the present disclosure, the probing unit includes at
20 least one quartz plate disposed on the top of the solar cell to allow full spectrum of incident wavelengths on the solar cell. The probing unit further includes at least one flexure unit disposed on the top surface of the solar cell configured to establish uniform and continuous contact.

[027] In another aspect of the present disclosure, the solar cell includes at least
25 two quartz plate disposed on a top and a bottom surface of the solar cell configured to allow full spectrum of incident wavelengths on the solar cell. The solar cell further includes a first flexure unit configured to establish uniform and continuous contact with top surface of the solar cell. Furthermore, the solar cell includes a second flexure unit configured to establish contact with the rear
30 surface of the solar cell.

[028] In another aspect of the present disclosure, the probing unit includes at least two parallel horizontal bars connected using a plurality of connectors to establish contact with the solar cell to enable the voltage and current measurement.

[029] In another aspect of the present disclosure, the plurality of connectors is one of: “arc”-shaped connectors and “Z”-shaped connectors.

[030] In another aspect of the present disclosure, the light mixer for solar cell electrical characteristics measurement system. The light mixer includes a housing, an array of light emitting diodes (LEDs), a four-layer LED panel and a mirror chamber. The mounting plate is horizontally placed at a bottom of the housing. The array of LEDs are disposed on the mounting plate, wherein the array of LEDs configured to produce a first light spectrum in a vertically upward direction. The four layer LED panel is disposed on top of the housing to produce a second light spectrum in a vertically downward direction. The mirror chamber is vertically placed over the mounting plate and disposed in between the mounting plate and the four-layer LED panel, the mirror chamber configured to receive light via array of LEDs and receive further light from the four-layer LED panel. The mirror chamber includes a plurality of mirrors facing inward to form a frustum shape mirror chamber to perform light mixing with high intensity for solar simulation. The heat sink is horizontally placed on top of LED panel to provide heat dissipation via coolant flowing through a copper cooling tube.

[031] In an aspect of the present disclosure, the plurality of mirrors are placed at a predefined angle to form the frustum mirror chamber.

[032] In an aspect of the present disclosure, the frustum mirror chamber is configured to reflect and blend the light received via the array of LEDs and the four-layer LED panel to produce a uniform light.

[033] In an aspect of the present disclosure, the light mixer provides simulated sunlight of the Sun Spectrum (AM1.5) with an intensity up to 1200W/m^2 .

[034] Various objects, features, aspects, and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[035] The illustrated embodiments of the subject matter will be understood by reference to the drawings, wherein like parts are designated by like numerals throughout. The following description is intended only by way of example and

simply illustrates certain selected embodiments of devices, systems, and methods that are consistent with the subject matter as claimed herein, wherein:

[036] **Figure 1** illustrates a pogo pin based traditional probing mechanism used for solar cell electrical characteristics measurement;

5 [037] **Figure 2a** illustrates an exploded view of vertical arrangement and layering of the light mixer's (SM1) various components, in accordance within an embodiment of the present disclosure;

[038] **figure 2(b)** and **2(c)** illustrates an exploded view and side view of the system, in accordance within an embodiment of the present disclosure;

10 [039] **Figure 3** illustrates the solar cell (SM2) and the probing unit (SM3) of the system, in accordance within an embodiment of the present invention;

[040] **Figure 4** illustrates the probing unit (SM3), in accordance within an embodiment of the present disclosure;

15 [041] **Figure 5** illustrates a flexure-based probing unit (SM3) designed for precision electrical contact with the solar cell (SM2);

[042] **Figure 6** illustrates the deviation of the light measured in a 10x10 matrix of 20cm x 20cm sample size in accordance within an embodiment of the present disclosure; and

20 [043] **Figure 7** illustrates the overlaid spectral measurements taken across a 10x10 matrix, in accordance within an embodiment of the present disclosure.

[044] The figures depict embodiments of the present subject matter for the purposes of illustration only. A person skilled in the art will easily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the disclosure described herein.

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DETAILED DESCRIPTION

[045] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the disclosure. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

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[046] While the embodiments of the disclosure are subject to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the figures and will be described below. It should be understood, however, that it is not intended to limit the disclosure to the particular forms disclosed, but on the contrary, the disclosure is to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure.

[047] The terms “comprises”, “comprising”, or any other variations thereof used in the disclosure, are intended to cover a non-exclusive inclusion, such that a device, system, or assembly that comprises a list of components does not include only those components but may include other components not expressly listed or inherent to such system, or assembly, or device. In other words, one or more elements in a system or device preceded by “comprises... a” does not, without more constraints, preclude the existence of other elements or additional elements in the system or device.

[048] **Figure 1** illustrates a pogo pin (201) based traditional probing mechanism used for solar cell electrical characteristics measurement. In order to obviate from the drawbacks of the existing system, present invention provides a comprehensive system comprising of LED based light mixer (SM1) configured to generate light spectrum close to AM 1.5 sun spectrum, wherein group of LED are employed to provide solar simulation. The LEDs are generating light spectrum of different wavelengths, in other words, an array of LEDs are configured to emit light spectrum of different wavelength to mimic / simulate natural sunlight.

[049] For better understanding, one or more embodiments of the present invention shall be described with respect to the earlier-mentioned drawings.

[050] The present invention introduces a comprehensive system for solar cell electrical characteristics measurement which includes: a light mixer (SM1) configured to simulate sunlight illumination. The system further includes a solar cell (SM2) configured to receive the simulated sunlight. Furthermore, the system includes a probing unit (SM3) for probing the solar cell for electrical characteristics measurement.

Exemplary embodiments of the invention

[051] **Figure 2a** illustrates the exploded view of vertical arrangement and layering of the light mixer's (SM1) various components, in accordance within an embodiment of the present disclosure. The figure 2(a) illustrates a LED Panel (102) configured for providing simulated sunlight. Said LED panel (102) incorporates more than 200 LEDs of capable of emitting the light spectrum of assorting wavelengths, close to natural sunlight spectrum (AM1.5 sun spectrums).

10 [052] Furthermore, **figure 2(b)** and **2(c)** illustrates an exploded view and side view of the system, in accordance within an embodiment of the present disclosure. The system may be one of monofacial cell or bifacial cell. The system includes of the light mixer (SM1), the solar cell (SM2), and the probing unit (SM3). The light mixer (SM1) is configured to simulate sunlight illumination.

15 The solar cell (SM2) is configured to receive the simulated sunlight. The probing unit (SM3) is used for probing the solar cell for electrical characteristics measurement.

[053] The light mixer (SM1) includes a housing, a mounting plate (102), an array of Light Emitting Diodes (LEDs) (104), a four layer LED panel (105), a mirror chamber (110) and at least one heat sink (108). The light mixer (SM1) may be configured to provide simulated sunlight of the Sun Spectrum (AM1.5) with an intensity up to 1200W/m^2 . The light mixer (SM1) mixes light with high intensity and efficiency. This light mixer (SM1) bypasses the need for additional optical filters, reducing light absorption losses and improving overall system efficiency. The light mixer (SM1) addresses the limitations of both Xenon and traditional LED-based simulators, by offering a more compact, efficient, and reliable solution for solar simulation.

25 [054] As can be observed from the figures 2(b) and 2(c), the mounting plate (102) is horizontally placed at the bottom of the housing. The LEDs (104) are disposed on the mounting plate (102). The LEDs produce a first light spectrum in a vertically upward direction. The four-layer LED panel (105) is disposed on top of the housing to produce a second light spectrum in a vertically downward direction.

[055] The mirror chamber (110) is vertically placed over the mounting plate (102) and disposed in between the mounting plate (102) and the four-layer LED panel (105). The mirror chamber (110) is configured to receive light via the LEDs (104) and receive further light from the four-layer LED panel (105). The mirror chamber (110) further includes a number of mirrors facing inward to form a frustum shape mirror chamber to perform light mixing with high intensity for solar simulation. The number of mirrors may be placed at a predefined angle to form the frustum mirror chamber (110). The frustum shape of the chamber (110), resembles a truncated cone or pyramid, it is designed to facilitate light mixing through internal reflections.

[056] Further, the mirror arrangement in the shape of a frustum helps mixing light within the frustum chamber (110) and does not require additional filters of the longer light path for the mixing. This gives a significant advantage of light intensity not being absorbed by the filters. The chamber (110) is configured to reflect and blend the light received via the array of LEDs (104) and the four-layer LED panel (105) to produce a uniform light. The heat sink (108) is horizontally placed on top of LED panel (105) to provide heat dissipation via coolant flowing through a copper cooling tube (106). The cooling tube (106) may be of a zig-zag shape.

[057] The heat sinks (108) is configured to cool down LEDs (104) for reliable operation which is also required for the Class AAA standards of temporal stability. In an embodiment of the present system, water-based cooling is used however other conventional cooling methods can also be employed for better thermal performance. The zig-zag shaped copper cooling tube (106) may be configured on this heatsink (108), however this tube, facilitates active cooling by carrying water coolant to remove heat from the heat sink.

[058] **Figure 3** illustrates the solar cell (SM2) and the probing unit (SM3) of the system, in accordance within an embodiment of the present invention. The solar cell (SM2) has been depicted in the figure 3(a) and the probing unit (SM3) has been depicted in the figure 3(b). The probing unit (SM3) includes at least one quartz plate (302) disposed on the top of the solar cell (SM2) to allow full spectrum of incident wavelengths on the solar cell (SM2). The incident wavelengths may range from 300 nm to 1200 nm. Further, the probing unit

(SM3) includes at least one flexure unit (304) disposed on the top surface of the solar cell (SM2) configured to establish uniform and continuous contact.

[059] The solar cell (SM2) includes at least two quartz plate (302, 308) disposed on top and bottom surface of the solar cell (SM2). The two quartz plate (302, 308) have been depicted in the figure 3(c). The quartz plate (302, 308) are configured to allow full spectrum of incident wavelengths on the solar cell (SM2). The solar cell (SM2) further includes the first flexure unit (304) configured to establish uniform and continuous contact with top surface of the solar cell (SM2). Furthermore, the solar cell (SM2) includes the second flexure unit (306) configured to establish contact with the rear surface of the solar cell (SM2). Additionally, at the bottom, a Gold plated vacuum chuck (310) enables proper rear contact for solar cell (SM3) with high conductivity. The gold plated vacuum chuck (310) has been depicted in the figure 3(d).

[060] **Figure 4** illustrates the probing unit (SM3), in accordance within an embodiment of the present disclosure. The probing unit (SM3) may be configured as a contact element along the length of a solar cell bus bar to measure electrical characteristics under illumination. Further, as can be observed from the figure 4, the probing unit (SM3) further includes at least two parallel horizontal bars (402, 404). The horizontal bars (402, 404) are connected using a number of connectors to establish contact with the solar cell (SM2) to enable the voltage and current measurement. The horizontal bars (402, 404) will correspond exactly to the number of busbars present in the solar cell (SM2) being measured. The number of connectors may be one of “arc”-shaped connectors (SM5) and “Z”-shaped connectors (SM6). The “arc” shaped connectors (SM5) have been depicted in the figure 4(a) and the “Z”-shaped connectors (SM6) have been depicted in the figure 4(b). The two parallel horizontal bars (402, 404) creates a flexible contact element with reduced shadowing effect and overall height.

[061] In an embodiment of the invention, the probing unit (SM3) can be plated with materials selected from group of highly conductive materials. The highly conductive materials may include gold. The gold-plated probing unit (SM3) may be designed to cater the issues like potential gradients and shadowing effects and provides structural integrity and eliminate the need for excessive height in the setup. The reduced height of the probing unit (SM3) prevents casting shadows on the solar cell (SM2) when exposed of simulated sunlight under light mixer,

thereby the system ensures that more light reaches the active surface, resulting in improved fill factor and more accurate IV (current-voltage) measurements.

[062] When the said probing unit is lowered onto the solar cell, the gold-plated contacts gently and uniformly touch the busbars with continuous electrical contact along the entire length of the busbar. A flat-bottomed design of the probing unit (SM3) ensures consistent pressure across the contact points, which eliminates the voltage gradients typically caused by uneven contact in traditional setups. This uniformity allows for continuous and stable probing of current and voltage, ensuring accurate readings without the fluctuations that can distort the IV curve.

[063] Figure 5 illustrates the flexure-based probing unit (SM3) designed for precision electrical contact with the solar cell (SM2), in accordance within an embodiment of the present disclosure. The solar cell (SM2) is positioned securely on a support platform, which ensures stable placement during the probing operation. A set of metallic probe arms 502, designed with compliant flexures, are aligned above the solar cell (SM2). The probe arms 502 are connected to contact fingers that establish the electrical connection with the busbars and fingers of the solar cell (SM2). This mechanism allows for low-force, high-reliability contact for IV measurements or characterization of solar cell (SM2), and the flexure design minimizes mechanical stress, thus protecting fragile wafer structures during repeated probing cycles.

[064] Figure 6 shows the deviation of the light measured in a 10x10 matrix of 20cm x 20cm sample size, in accordance within an embodiment of the present disclosure. The measurement was performed using a standard spectrometer positioned 10 cm away from the LED panel. The intensity readings were taken at equidistant points across the same plane, and the deviations from the maximum intensity were plotted. The maximum deviation observed was 4%.

[065] Figure 7 shows the overlaid spectral measurements taken across a 10x10 matrix, in accordance within an embodiment of the present disclosure. The spectral measurements may be 100 in total. The measurements are taken both with and without the mirror. The measurements illustrate the spectral consistency and effect of the mirror.

Advantages-

- The present disclosure ensures accurate simulation of sunlight both with high intensity and spectral uniformity.
- The present disclosure prevents shadowing effects on the solar cell during measurement acquisition.
- The present disclosure prevents potential gradients across the busbars during IV measurements.

[066] It should be noted that the description and figures merely illustrate the principles of the present subject matter. It should be appreciated by those skilled in the art that conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present subject matter. It should also be appreciated by those skilled in the art by devising various systems that, although not explicitly described or shown herein, embody the principles of the present subject matter and are included within its spirit and scope. Furthermore, all examples recited herein are principally intended expressly to be for pedagogical purposes to aid the reader in understanding the principles of the present subject matter and the concepts contributed by the inventor(s) to further the art and are to be construed as being without limitation to such specifically recited examples and conditions.

The novel features which are believed to be characteristic of the present subject matter, both as to its organization and method of operation, together with further objects and advantages will be better understood from the following description when considered in connection with the accompanying figures.

[067] Although embodiments for the present subject matter have been described in language specific to package features, it is to be understood that the present subject matter is not necessarily limited to the specific features described. Rather, the specific features and methods are disclosed as embodiments for the present subject matter. Numerous modifications and adaptations of the system/device of the present invention will be apparent to those skilled in the art, and thus it is intended by the appended claims to cover all such modifications and adaptations which fall within the scope of the present subject matter.

We Claim:

1. A system for solar cell electrical characteristics measurement, comprising:
 - a light mixer (SM1) configured to simulate sunlight illumination;
 - a solar cell (SM2) configured to receive the simulated sunlight; and
 - 5 – a probing unit (SM3) for probing the solar cell for electrical characteristics measurement.
2. The system as claimed in claim 1, wherein said solar cell may be one of monofacial cell and bifacial cell.
3. The system as claimed in claim 2, wherein the probing unit (SM3) comprises:
 - 10 – at least one quartz plate (302) disposed on the top of the solar cell (SM2) to allow full spectrum of incident wavelengths on the solar cell (SM2); and
 - at least one flexure unit (304) disposed on the top surface of the solar cell (SM2) configured to establish uniform and continuous contact.
- 15 4. The system as claimed in claim 2, wherein the solar cell (SM2) comprises:
 - at least two quartz plate (302, 308) disposed on a top and a bottom surface of the solar cell (SM2) configured to allow full spectrum of incident wavelengths on the solar cell (SM2);
 - a first flexure unit (304) configured to establish uniform and
 - 20 continuous contact with top surface of the solar cell (SM2); and
 - a second flexure unit (306) configured to establish contact with the rear surface of the solar cell (SM2).
5. The system as claimed in claim 1, wherein the probing unit (SM3) comprises at least two parallel horizontal bars (402, 404) connected using a plurality of
- 25 connectors to establish contact with the solar cell (SM2) to enable the voltage and current measurement.
6. The system as claimed in claim 5, wherein the plurality of connectors is one of: “arc”-shaped connectors (SM5) and “Z”-shaped connectors (SM6).
7. A light mixer (SM1) for solar cell electrical characteristics measurement
- 30 system, the light mixture (SM1) comprising:
 - a housing;
 - a mounting plate (102) horizontally placed at a bottom of the housing;

an array of light emitting diodes (LEDs) (104) disposed on the mounting plate (102), wherein the array of LEDs (104) configured to produce a first light spectrum in a vertically upward direction;

5 a four-layer LED panel (105) disposed on top of the housing to produce a second light spectrum in a vertically downward direction;

a mirror chamber (110) vertically placed over mounting plate (102) and disposed in between the mounting plate (102) and the four-layer LED panel (105), the mirror chamber (110) configured to receive light via array of light emitting diodes (LEDs) (104) and receive further light from the four-layer LED panel (105),
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wherein the mirror chamber (110) comprises a plurality of mirrors facing inward to form a frustum shape mirror chamber to perform light mixing with high intensity for solar simulation;

at least one heat sink (108) horizontally placed on top of LED panel (105) to provide heat dissipation via coolant flowing through a copper cooling tube (106).
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8. The light mixture (SM1) as claimed in claim 7, wherein the plurality of mirrors are placed at a predefined angle to form the frustum mirror chamber (110).

20 9. The light mixture (SM1) as claimed in claim 7, wherein the frustum mirror chamber (110) is configured to reflect and blend the light received via the array of LEDs and the four-layer LED panel (105) to produce a uniform light.

10. The light mixture (SM1) as claimed in claim 7, wherein the light mixer (SM1) provides simulated sunlight of the Sun Spectrum (AM1.5) with an intensity up to 1200W/m².
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Dated this 07th day of August, 2025

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35

ABSTRACT

“SYSTEM FOR SOLAR CELL ELECTRICAL CHARACTERISTICS MEASUREMENT”

5 The present subject matter discloses a system for solar cell electrical characteristics
measurement. The system comprises a light mixer (SM1) for solar simulation, a solar
cell (SM2) under solar illumination, and a probing unit (SM3) to measure electrical
characteristics of said solar cell. The light mixer (SM1) utilizes a frustum-shaped
mirror chamber (110) and LED light sources (104,105) to generate highly uniform
10 and intense simulated sunlight. The probing unit (SM3) is designed to minimize
shadowing and ensures reliable electrical contact for accurate electrical
measurements of solar cell (SM2).

(FIG. 2 and 3)